

Power BI Project 1 | Car Sales Analysis

①

> To build a dashboard (3 requirements)

① Sales Overview :-

- * YTD total Sales
- * MTD total Sales
- * YOY Growth in Total Sales
- * Difference between YTD Sales & Previous YTD Sales

② Average Price Analysis :-

- * YTD Avg Price
- * MTD Avg Price
- * YOY Growth in Avg Price
- * Difference between YTD Price & Previous YTD Price

③ Cars Sold Metrics :-

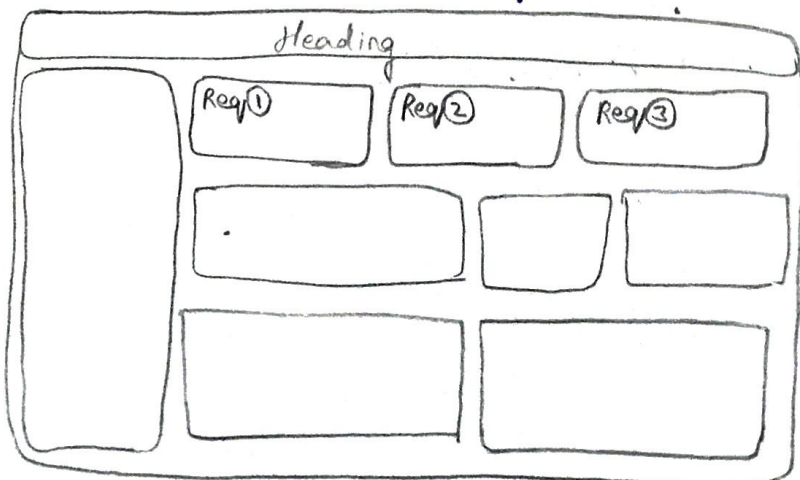
- * YTD Cars sold
- * MTD Cars sold
- * YOY Growth in Cars sold
- * Difference between YTD Cars Sold & PYTD Cars Sold.

○ Excel Worksheet → Car Sales

○ Power BI

Layout → Dashboard

Created using
MS Powerpoint

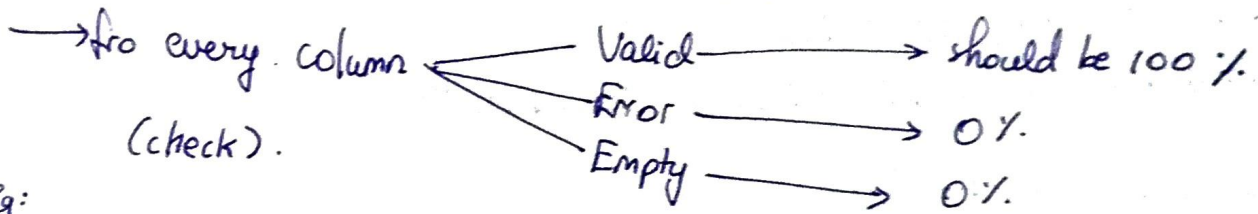


Procedure

(2)

- * Oper Power BI app → New sheet → Import
 - * Check the data type if excel → import data from Excel
= SQL server / Other source
 - * Now import the Excel "Car Sales" → Oper → Load Data.
 - * Go to Table view [Left Pane]
- ① Check for data quality!

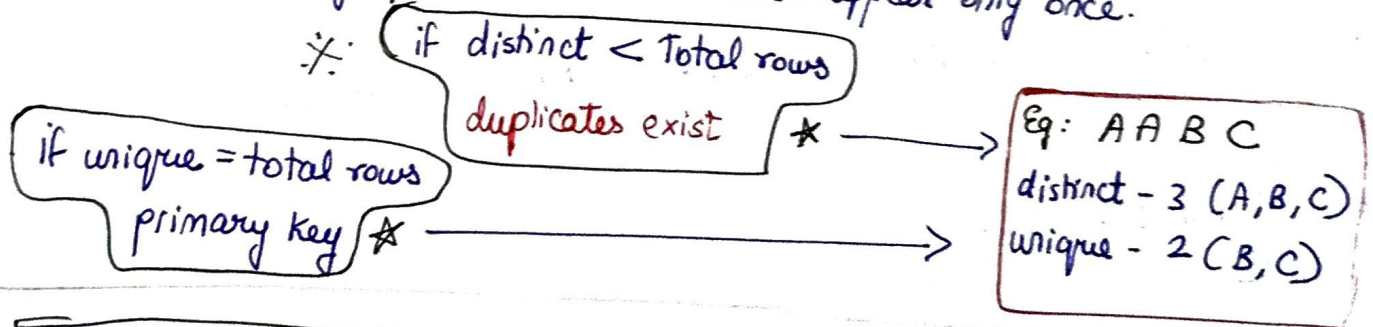
- * Transform data → Analyse data
- * View Table → Check → Column quality
Column distribution.



Eg:
1000 distinct / 1000 unique.

each row → 1000 different values
↳ some values may repeat

has 1000 values
all values appear only once.



② Data Analyse

- * Mandatory columns should have 0% null value.

Eg: Date of Sales
Price of Cars
User Name

③ Data Cleaning.

→ purpose

→ To correct errors

:- spelling.

:- Numerical format

- * Right click any column header
- * List of all the unique values pop up
- * Check all them for any error →

if Yes

→ First copy the cell value containing error

→ Right click > Replace > Correct it > ok

* changes applied

① YTD, MTY, YOY Sales → All depend on the calendar

∴ We create a new Table

Calendar Table

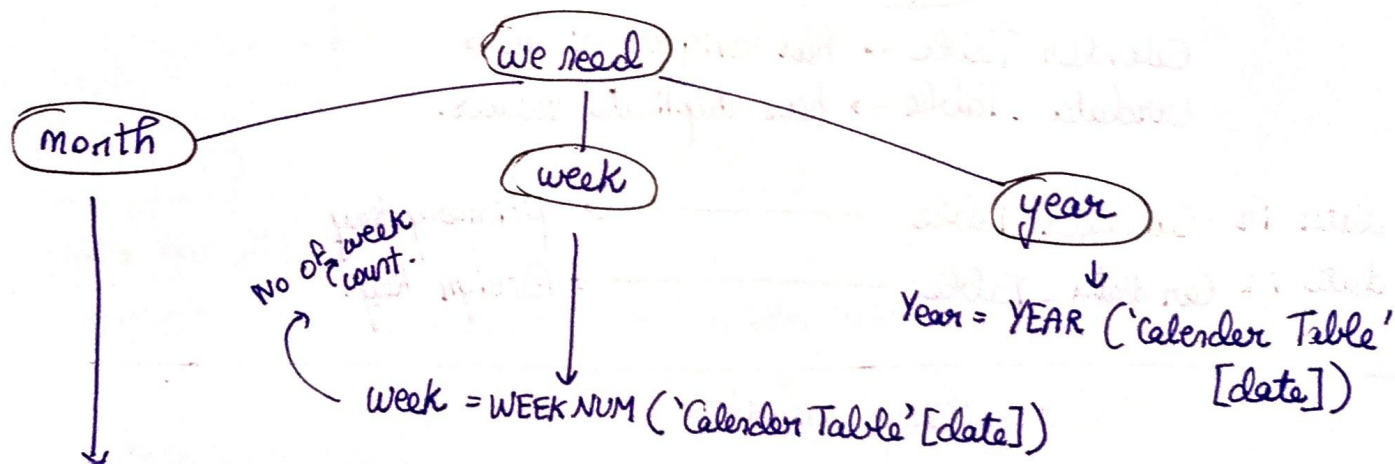
→ To perform different operations on dates

* Home → Calculation Tab → New Table → Rename as "Calendar Table"

* $\text{Calendar Table} = \text{CALENDAR}(\text{MIN}(\text{cardata_Table}[\text{date}] - \text{MAX}(\text{cardata_Table}[\text{date}]))$

Now we get a sorted date column.

Each row - carry unique value



Month = $\text{Format}(\text{'Calendar Table' [date]}, \text{"MMMM"})$

↳ Ques 1:- Why FORMAT? and not Month function?

Month (data) → return numerical value (1, 2, -12)

Format (data) → returns month name [July, May]

Q2) Why "MMMM" and not "MM"?

↳ mm → Month number { 05 → May
09 → September

↳ mmmm → represents the word type to display month.
→ May, June, July

Now Calendar Table (seen as)

Date	Year	Month	Week

Now Calendar Table & Carddata-Table are two independent Tables → To apply relationship b/w them.

↳ Model view → (date) is common on both Tables
* Drag from Calendar Table → drop in Main Table
* Relationship formed → [1 to many]

Why?

Calendar Table → has unique values
Carddata-Table → has duplicate values.

* date in Calendar Table → primary key
* date in Carddata-Table → foreign key.

Start Building Visualise

> Report View → Format Report → Canvas → Canvas Background →
Image → Add from storage → Open.

> Transperency → 100% → 0%.

> Vertical Alignment → Top → middle → Change

① To find YTD Total Sales

- > click drop-down carddata-table in right pane
- > Using both the Tables create it
- > Click anywhere [right click in carddata Table] > New Measure

$$\text{YTD Total Sales} = \text{Total YTD}(\text{SUM}(\text{cardata Table} [\text{Price} (\$)]), \text{'Calendar Table' [Date]})$$

* Automatically Takes the latest year.

> Click ok > saved in right pane.

> Visualization → Select card > Drag & Drop YTD Table Sales into the canvas

- ① > Format Visual → General → Effect → Background (off)
- ② > Format visual → Multicard layout → Background (off)
- ③ > Format visual → Cards → Background (off)
- ④ > Callout → Value → Desired values And font color (grey)
- ⑤ > Callout → Label → off
- ⑥ > General → Title → On → Give title as "YTD data Sales"
- ⑦ > Callout → Value → "Center Align" "Center align"
- ⑧ > Adjust total size & place it in the canvas "position"
- ⑨ > Select "YTD Table Sales" in the right pane → Measure Tools → Click \$ symbol → Auto x → 1/2 decimal place

② To find YOY & Diff of YTD & PYTD

> Again → New Measure

$$\text{PYTD Total Sales} = \text{CALCULATE}(\text{SUM}(\text{cardata-Table} [\text{Price} (\$)]), \text{SAMEPERIODLASTYEAR}(\text{'Calendar Table' [Date]}))$$

Click Enter

> Again → New Measure

$$\text{Sales Difference} = [\text{YTD Total Sales}] - [\text{PYTD Total Sales}]$$

Click Enter.

> New Card → for sales difference

> Now seen as

71M
Sales Diff

> Select the prev card → Format Painter → click Here. → Give currency & 1 decimal → Enter.

> Format for font size.

Conditional format

BUT Now

Value → profit → green
Value → loss → red] dynamic

> New Measure → Sales Diff colour = IF ([Sales Difference] > 0, "Green", "Red")

> select Card → Format Visual → Callout Value → fx → Rules → Sales. Diff
Colour → IF value is Green then ☐ choose green
> New rule - "Red" the ☐ choose red
⇒ ok.

To Find YOY Growth in Total Sales

⇒ New Measure

$$\text{YOY Sales Growth} = \frac{[\text{Sales Difference}]}{[\text{PYTD Total Sales}]}$$

> Copy & Paste the Card used for Diff b/w YTD Price & PYTD Price
> Field → change the YOY Sales Growth (or)
> Drag & Drop the "YOY Sales Growth" to the new card.
> Visualization → Format → Visual → Callout Value → fx → Modify → Save
> Change it to % & upto 2 decimal

④ MTD Total Sales

> New Measure → MTD Total Sales = TOTAL MTD (SUM ('Calendar Table'

① [Price (\$)], 'Calendar Table' [Date])

> save it.

② Measure → (create a new one)

MTD KPI = CONCATENATE ("MTD Total Sales : ", FORMAT ([MTD Total Sales] / 1000000, "\$0.00M"))

↳ why 2 measures for this field?

→ To display the result with the string "MTD Total Sales", and for correct formatting of the value (ie) To display it in the million type.

→ This is mainly to join the String & the calculation

Now ① Requirement (Sales Overview) is completed

Avg Price Analysis

YTD Avg Price

MTD Avg Price

Yoy Growth in Avg Price

Diff b/w YTD Price & PYTD Price.

→ important fields :

'carData' table [Price (\$)]
'Calendar Table' [date]

* To follow the same formatting as requirement - 1

① YTD Avg Price = TOTAL YTD ([Avg Price], 'Calendar Table' [date])

$$\text{Avg Price} = \frac{\text{SUM ('carData' - Table [Price ($)])}}{\text{COUNT ('carData' - Table [car-id])}}$$

why car-id Column?

↳ It is a not null column & its used as a parameter to count the cars.

↳ This is a primary key, therefore one car can be purchased 1

② ~~YTD~~ YTD Price - PYTD Price

↳ Measure (create new)

~~PYTD Avg Price = CALCULATE ([Avg Price], 'Calendar Table' [date])~~

⇒ PYTD = CALCULATE ([Avg Price], SAME PERIOD LAST YEAR
('Calendar Table' [Date]))

⇒ New Measure →

Avg Diff... Avg Price Diff = [YTD Avg Price] - [PYTD Avg Sales]

③ YOY Growth in Avg Price

⇒ New Measure →

YOY Avg Price Growth = $\frac{[Avg Price Diff]}{[PYTD Avg Sales]}$

For dynamic colour change :-

⇒ New Measure →

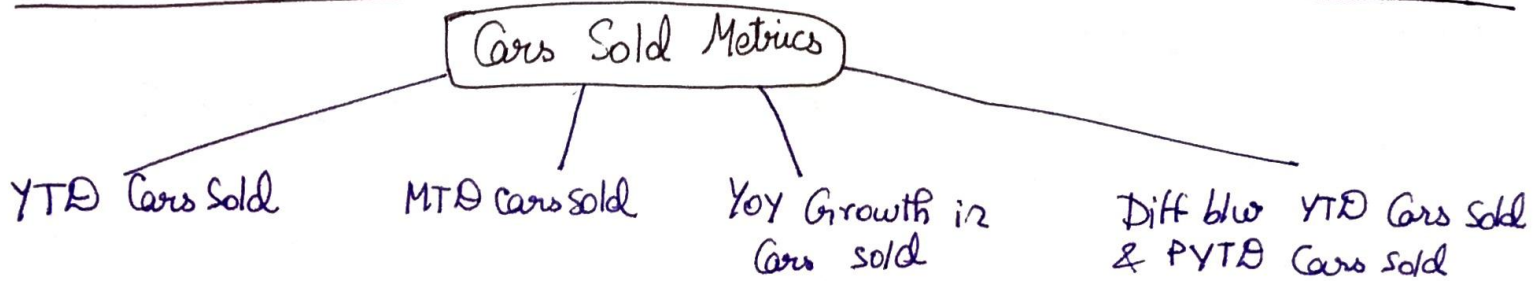
Avg Price Colocur = IF([Avg Price Diff] > 0, "Green", "Red")

④ MTD Avg Price

MTD Avg Price = TOTALMTD([Avg Price], 'Calendar Table' [Date])

MTD Avg KPI = CONCATENATE("MTD Avg Price : ", FORMAT[MTD Avg Price] / 1000, "\$0.001").

Requirement ② Completed



① YTD Cars Sold = TOTALYTD(COUNT('carsdata_Table'[car-id], 'Calendar Table'[Date]))

② Diff b/w YTD Cars Sold & PYTD Cars Sold.

PYTD Cars Sold = CALCULATE(COUNT(CarData-Table[Car id]),
SAME PERIOD LAST YEAR('Calendar Table'[Date]))

CARS Sold diff = $\frac{[YTD \text{ Cars Sold}] - [PYTD \text{ Cars Sold}]}{[YTD \text{ Cars Sold}]}$

Cars → ③ YOY Growth in Cars Sold

→ New Measure →

YOY Car sold growth = $\frac{[Cars \text{ Sold diff}]}{[YTD \text{ Cars Sold}]}$

For dynamic colour change :-

→ New Measure → Car Sold colour = IF (CarData-Table [Cars sold diff] > 0,
"Green", "Red")

④ MTD Cars Sold.

MTD Cars Sold = TOTALMTD (COUNT (CarData-Table [Car id]),
'Calendar Table'[date])

MTD Car KPI = CONCATENATE ("MTD Car Sold : ", FORMAT
[MTD Car sold] / 1000, "\$ 0.00k")

Requirement ③ Completed

* CHARTS REQUIREMENT (6)

- ① YTD Sales Weekly Trend - line chart
- ② YTD Total sales by Body Style - Pie chart
- ③ YTD Total Sales by Colour - Pie chart
- ④ YTD Cars Sold by Dealer Region - map chart (geographically)
- ⑤ Compy-Wise Sales Trend in Grid Form - Tabular grid
- ⑥ Details Grid Showing All Car Sales Info - Grid which present all relevant info.

① The YTD Sales Weekly Trend

> Choose Area Chart > Align it in placeholder

X axis $\longrightarrow$ Week $\longrightarrow$ from calendar Table

Y axis $\longrightarrow$ Sum of Price $\longrightarrow$ from cardata - Table

> Filter $\rightarrow$ Year (drag & drop).

$\rightarrow$ Basic Filtering $\rightarrow$ Choose (2023) as the latest year.

> General $\rightarrow$ effects $\rightarrow$ Background off $\rightarrow$ Title (give as YTD Sales Weekly Trend) $\rightarrow$ Xaxis $\rightarrow$ max point set it to 54 [as there are only 54 weeks]

* For marking the Max point :-

② Measure $\rightarrow$ Max point on line graph = $IF (MAX (SELECTED ('calendar Table' [week])), [Total Sales]) = [Total Sales], (MAX (ALL SELECTED ('calendar Table' [week])), BLANK ()))$

① Measure $\rightarrow$ for Total sales = $SUM ('cardata-table' [Price ($)])$

* Marker $\rightarrow$ on $\rightarrow$ Series [MAX POINT] $\rightarrow$ Colour - Choose [PINK / YELLOW]

Turn on the label for MAX POINT

Options $\rightarrow$ Position $\rightarrow$ All $\rightarrow$ Value (off)
 $\downarrow$
 * sum of Price (\$) $\uparrow$

Data labels $\rightarrow$ Series $\rightarrow$ Max Point $\rightarrow$ Upet the result.

② YTD Total Sales by Body Style.] donut charts.

③ YTD Total Sales by Colour.

$\rightarrow$ choose the choose $\rightarrow$ donut chart,

$\rightarrow$ align it in the place holder.

① Body Style $\rightarrow$ drop it in legend
 YTD Total sales $\rightarrow$ Values.

② Body Colour $\rightarrow$ drop it in legend
 YTD Total sales $\rightarrow$ Values.

$\rightarrow$ Turn ON legends..

$\rightarrow$ Colour change $\rightarrow$ Slices $\rightarrow$ Choose the colours & ok.

$\rightarrow$ Add Proper title.

④ YTD Cars Sold by Dealer Region

- Use of map graph.
- Drag & drop the map graph in the placeholder
- Region → Add in the location
- YTD Cars Sold → Add in the Bubble size
- Formatting :-
 - Category Label → ON
 - * Map Settings → Dark
 - * Bubbles → Size → Range scaling → +10
 - * If you want change the Bubble colour.
 - * Turn off the Background.

⑤ Company Wise Sales Trend

- Use the Table graph.
- In Columns add :-
 - Company
 - YTD Avg Price
 - YTD Car Sold
 - YTD Total Sales
 - YTD Total Sales (grand total) → show value as →
- Conditional formatting → ~~Bar graphs~~ [%. of grand Totals]
- Formatting → Minimal [Style]
 - Background [off]
 - gridlines → Vertical ✓ / Horizontal x
 - Text colour → [grey white]
 - Totals → off..
 - Column header — Black
 - Column header Text → grey white.
 - Center alignment

for Car sold & Total Sales → Turn on Data Bars → in Cond. formatting

- +ve Bars - pink
- ve Bars - Baby Blue. >ok.
- Add Title of the Table

- Next Add Header of the Dashboard.
- Add Name on the right top corner → for authentication & ownership
- Add navigation Buttons

Insert → Buttons → Navigation Buttons -
Format it → font & colour.

- Add Filter Buttons →

Visualizer → Slicer → Click → Align it in Placeholder
Field → Add Body Style → Format - Background - Black -
- Button - White / Grey
- Text - white

Slicer setting - Drop down

Selection → Multi select
Show select } all tickboxes.

3 Filter Buttons → Body Style
→ Dealer - Name
→ Transmission.

Add Another sheet

Name 1st sheet as Overview / Metrics / Dashboard
Name 2nd sheet as Table View

Delete all the line graph
2 donut chart
Map chart
Table chart

Retain → The metrics on top
The controls on left Panel

→ Add Metrics & place it & align it in placeholder.

→ Columns drag & drop :-

{	* Car-id	* Customer Name	* Company	* Model
	* Date	* Dealer Name	* Colour	* Total sales

- Turn off Background → Text in grey white
- Turn off column header → Column header text → in Pink
- Turn off horizontal grid lines
- Total → highlight
- Total sales → Conditional formatting → Add data Bars.